# CRITERION FOR A SCHEME TO REPRESENT THE QUOTIENT SHEAF U/R

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*Allowed standard prerequisites (editorial).*

- Definition definition-quotient-sheaf (canonical tag 02VG). cited at statement line 3983. U/R is the tau-sheaf quotient; its universal property gives the factorization.

*Source reference.* Original definition 02VG: sources/stacks-chow/tags.txt, source lines 3767–3767. Cited at statement line 3983: the lemma is phrased in the situation of this definition; load-bearing.

**Lemma 0.2.** *In the situation of Definition 02VG. Assume there is a scheme $M$, and a morphism $U \to M$ such that*

1. *the morphism $U \to M$ equalizes $s, t$,*
2. *the morphism $U \to M$ induces a surjection of sheaves $h_U \to h_M$ in the $\tau$-topology, and*
3. *the induced map $(t, s) : R \to U \times_M U$ induces a surjection of sheaves $h_R \to h_{U \times_M U}$ in the $\tau$-topology.*

*In this case $M$ represents the quotient sheaf $U/R$.*

*Proof.* Condition (1) says that $h_U \to h_M$ factors through $U/R$. Condition (2) says that $U/R \to h_M$ is surjective as a map of sheaves. Condition (3) says that $U/R \to h_M$ is injective as a map of sheaves. Hence the lemma follows. □

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